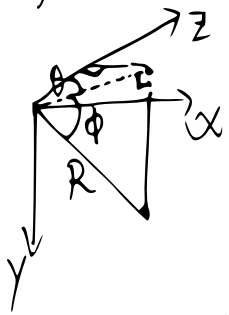
$\therefore$ if $\lambda \le 0$, X can't be projected onto image.

1. pinhole projection
   (perspective projection, rectilinear projection)

1.3 $(x, y, z) \to (u, v)$



solve: $\begin{pmatrix} u \\ v \\ f \end{pmatrix} = f \begin{pmatrix} \frac{x}{z} \\ \frac{y}{z} \\ 1 \end{pmatrix}$

$\therefore \begin{pmatrix} u \\ v \end{pmatrix} = f \begin{pmatrix} \frac{x}{z} \\ \frac{y}{z} \end{pmatrix}$, f is focal length

1.4 $(u, v) \to (x, y, z)$

solve: $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} u \\ v \\ f \end{pmatrix}$

0.5.3

$\therefore$ affine homography is a special homography.

0.5.4

$\therefore$ cameras $K_1(R\ t)$, $K_2(R, t)$ can represent image affine trans, i.e., affine homography.

$\therefore$ cameras simplify to $K_1(I\ 0)$, $K_2(I\ 0)$ when representing affine, or just K_1, K_2, or R_1, R_2 if abusing notation.

0.6 Given camera K, R, t,

$\therefore \lambda \begin{pmatrix} x \\ 1 \end{pmatrix} = K(R\ t) \begin{pmatrix} X \\ 1 \end{pmatrix} = K X_{cam}$

K is just affine

$\therefore \lambda = X_{cam}(3)$

$\boxed{\text{projection models}}$

0. camera

0.3 projectivity

given $K, R, t \Rightarrow \lambda \begin{pmatrix} x \\ 1 \end{pmatrix} = K(R, t) \begin{pmatrix} X \\ 1 \end{pmatrix}$

$x \in R^2$, $X \in R^3$

0.4 homography

$\lambda_1 \begin{pmatrix} x_1 \\ 1 \end{pmatrix} = K_1 R_1 X$, $\lambda_2 \begin{pmatrix} x_2 \\ 1 \end{pmatrix} = K_2 R_2 X$

$\therefore \lambda_2 \begin{pmatrix} x_2 \\ 1 \end{pmatrix} = K_2 R_2 \lambda_1 R_1^{-1} K_1^{-1} \begin{pmatrix} x_1 \\ 1 \end{pmatrix}$

$\therefore \lambda \begin{pmatrix} x_2 \\ 1 \end{pmatrix} = H_{1 \to 2} \begin{pmatrix} x_1 \\ 1 \end{pmatrix}$

0.4.3 if $K_1 = K_2 \Rightarrow H_{1 \to 2} = R_2 R_1^{-1}$

0.5 affine homography

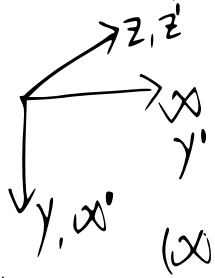
let $R_1 = R_2$ in 0.4. $\Rightarrow H_{1 \to 2} = K_2 K_1^{-1}$

here $H_{1 \to 2}$ must be affine, (see chain of H) and $\lambda = 1$

solve: $\begin{pmatrix} y \\ v \end{pmatrix} \leftarrow r^{-1}\begin{pmatrix} u \\ v \end{pmatrix}$, $v \leftarrow \pi - v$

$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} sv\,su \\ cv \\ sv\,cu \end{pmatrix}$

2.b spherical portrait projection



2.b.3 $(x, y, z) \to (u, v)$

solve: $\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} \leftarrow \begin{pmatrix} y \\ x \\ z \end{pmatrix}$, $\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} \leftarrow \left\|\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right\|^{-1}\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$

$\therefore u' = \tan^{-1}\left(\frac{x'}{z'}\right)$, $v' = \pi - \cos^{-1} y'$

$\therefore \begin{pmatrix} u' \\ v' \end{pmatrix} \leftarrow r\begin{pmatrix} u' \\ v' \end{pmatrix}$

$\therefore \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v' \\ u' \end{pmatrix}$ or $\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} -u' \\ v' \end{pmatrix}$

2.b.3.2: given $I = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$,

if $\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v' \\ u' \end{pmatrix} \Rightarrow$ projected I is $\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$

if $\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} -u' \\ v' \end{pmatrix} \Rightarrow$ projected $I = \begin{pmatrix} 5 & 3 & 1 \\ 6 & 4 & 2 \end{pmatrix}$

1.8 $(u, v) \to (\theta, \phi)$

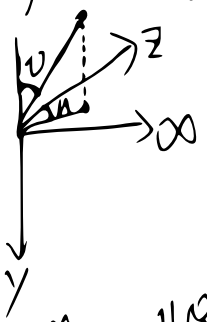
solve: $\begin{pmatrix} \theta \\ \phi \end{pmatrix} = \begin{pmatrix} \tan^{-1}(u, f) \\ \sin^{-1}\left(v / \left\|\begin{pmatrix} u \\ v \\ f \end{pmatrix}\right\|\right) \end{pmatrix}$

2. panoramic spherical projection

(equirectangular projection)

here equirectangular: equal-angle rectangular

2.3 $(x, y, z) \to (u, v)$



solve: $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \leftarrow \left\|\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right\|^{-1}\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\therefore u = r\tan^{-1}(x, z) = r\tan^{-1}\left(\frac{x}{z}\right)$

$v = r(\pi - \cos^{-1} y)$

2.4 $(u, v) \to (x, y, z)$

1.5 $(R, \theta, \phi) \to (x, y, z)$

θ: longitude, ϕ: latitude

yaw, pitch

$[-\pi, \pi]$, $[-\frac{\pi}{2}, \frac{\pi}{2}]$

solve: $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = R\begin{pmatrix} \cos\phi \sin\theta \\ \sin\phi \\ \cos\phi \cos\theta \end{pmatrix}$

1.6 $(x, y, z) \to (R, \theta, \phi)$

solve: $R \leftarrow \left\|\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right\|$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \leftarrow R^{-1}\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\therefore \begin{pmatrix} \theta \\ \phi \end{pmatrix} = \begin{pmatrix} \tan^{-1}(x, z) \\ \sin^{-1}(y) \end{pmatrix}$

1.7 $(\theta, \phi) \to (u, v)$

solve: 1.5, 1.3 $\Rightarrow \begin{pmatrix} u \\ v \end{pmatrix} = f\begin{pmatrix} \frac{x}{z} \\ \frac{y}{z} \end{pmatrix}$

$= f\begin{pmatrix} \tan\theta \\ \tan\phi / \cos\theta \end{pmatrix}$

4.3 $(x,y,z)\to(u,v)$

solve: $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow z^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$, $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow(1-t_3)\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\therefore z=(1-t_3)$, $\begin{pmatrix}u\\v\end{pmatrix}=\begin{pmatrix}x\\y\end{pmatrix}+\begin{pmatrix}t_1\\t_2\end{pmatrix}$

$\therefore \begin{pmatrix}u\\v\end{pmatrix}\leftarrow r\begin{pmatrix}u\\v\end{pmatrix}$

4.4 $\begin{pmatrix}u\\v\end{pmatrix}\to\begin{pmatrix}x\\y\\z\end{pmatrix}$

solve: $\begin{pmatrix}u\\v\end{pmatrix}\leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}$

$\therefore \begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}u\\v\end{pmatrix}-\begin{pmatrix}t_1\\t_2\end{pmatrix}$

$z=1-t_3$

4.5 sensor plane is at $z=1-t_3$

4.b plane portrait projection

4.b.3 $(x,y,z)\to(u,v)$

(similar to 2.b.3)

solve: $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow\begin{pmatrix}x\\y\\z\end{pmatrix}$, $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow z^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$, $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow(1-t_3)\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\therefore z=1-t_3$, $\begin{pmatrix}u\\v\end{pmatrix}=r\left(\begin{pmatrix}x\\y\end{pmatrix}+\begin{pmatrix}t_1\\t_2\end{pmatrix}\right)$

$\therefore \begin{pmatrix}u\\v\end{pmatrix}\leftarrow\begin{pmatrix}-u\\v\end{pmatrix}$

3.b cylindrical portrait projection

3.b.3 $(x,y,z)\to(u,v)$

(similar to 2.b.3)

solve: $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow\begin{pmatrix}y\\x\\z\end{pmatrix}$, $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow(x^2+z^2)^{-\frac{1}{2}}\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\therefore \begin{pmatrix}u\\v\end{pmatrix}=r\begin{pmatrix}\tan^{-1}\left(\frac{x}{z}\right)\\y\end{pmatrix}$, $\begin{pmatrix}u\\v\end{pmatrix}\leftarrow\begin{pmatrix}-u\\v\end{pmatrix}$

3.b.4 $(u,v)\to(x,y,z)$

solve: $\begin{pmatrix}u\\v\end{pmatrix}\leftarrow\begin{pmatrix}-u\\v\end{pmatrix}$, $\begin{pmatrix}u\\v\end{pmatrix}\leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}$

$\therefore \begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}y\\x\\z\end{pmatrix}$

4. plane projection

(modified pinhole projection: added parameter t ($\neq$ camera translation t))

4.2 $t=\begin{pmatrix}t_1\\t_2\\t_3\end{pmatrix}$,

t_1, t_2: offset

$1-t_3$: distance between world origin and plane.

2.b.4 $(u,v)\to(x,y,z)$

solve: $\begin{pmatrix}u'\\v'\end{pmatrix}\leftarrow\begin{pmatrix}-u\\v\end{pmatrix}$, $\begin{pmatrix}u'\\v'\end{pmatrix}\leftarrow r^{-1}\begin{pmatrix}u'\\v'\end{pmatrix}$,

$v'\leftarrow\pi-v'$

$\therefore \begin{pmatrix}x'\\y'\\z'\end{pmatrix}=\begin{pmatrix}s v' s u'\\c v'\\s v' c u'\end{pmatrix}$

$\therefore \begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}y'\\x'\\z'\end{pmatrix}$

3. cylindrical projection

3.3 $(x,y,z)\to(u,v)$

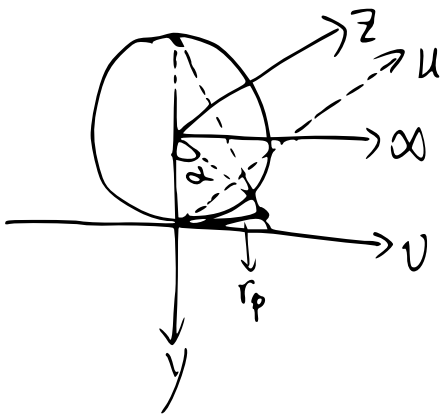
solve: $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow(x^2+z^2)^{-\frac{1}{2}}\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\therefore \begin{pmatrix}u\\v\end{pmatrix}=r\begin{pmatrix}\tan^{-1}(x,z)\\y\end{pmatrix}$

3.4 $(u,v)\to(x,y,z)$

solve: $\begin{pmatrix}u\\v\end{pmatrix}\leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}$, $\therefore \begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}s u\\v\\c u\end{pmatrix}$

6.3 $(x,y,z) \to (u,v)$



solve: $\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \left\|\begin{pmatrix}x\\y\\z\end{pmatrix}\right\|^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\begin{pmatrix}u\\v\end{pmatrix} = \begin{pmatrix}\tan^{-1}(x,z)\\ \pi - \cos^{-1} y\end{pmatrix}$

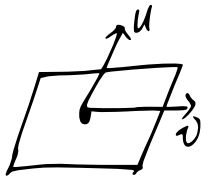
let $\alpha = \pi - v$

$\therefore r_{p_2} = \frac{r_p}{2} = \tan\frac{\alpha}{2} = \frac{\sin\alpha}{1+\cos\alpha} = \frac{sv}{1-cv}$

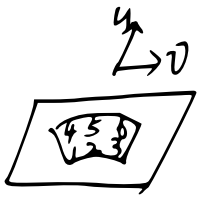
$\therefore \begin{pmatrix}u\\v\end{pmatrix} \leftarrow r_{p_2}\begin{pmatrix}cu\\su\end{pmatrix}$, $\therefore \begin{pmatrix}u\\v\end{pmatrix} \leftarrow r\begin{pmatrix}u\\v\end{pmatrix}$

6.3.2 $\therefore \begin{pmatrix}1&2&3\\4&5&6\end{pmatrix} \to \begin{pmatrix}1&2\\5&6\\3&4\end{pmatrix}$

6.4 $(u,v) \to (x,y,z)$



$\therefore$ for $I = \begin{pmatrix}1&2&3\\4&5&6\end{pmatrix} \Rightarrow \begin{pmatrix}1&4\\2&5\\3&6\end{pmatrix}$



5.4 $(u,v) \to (x,y,z)$

solve: $\begin{pmatrix}u\\v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}$

$\therefore \begin{pmatrix}u\\v\end{pmatrix} \leftarrow \begin{pmatrix}\tan^{-1}(v,u)\\ \|\begin{pmatrix}u\\v\end{pmatrix}\|\end{pmatrix}$

$v \leftarrow \pi - v$

$\therefore \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}sv\,su\\ cv\\ sv\,cu\end{pmatrix}$

5.5 fisheye sensor plane is at $Y=-1$.

6. stereographic projection

6.2 name meaning:

draw 3d sphere onto 2d plane

4.6.4 $(u,v) \to (x,y,z)$

solve: $\begin{pmatrix}u\\v\end{pmatrix} \leftarrow \begin{pmatrix}-u\\v\end{pmatrix}$, $\begin{pmatrix}u\\v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}$

$\therefore \begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}u\\v\end{pmatrix} - \begin{pmatrix}t_1\\t_2\end{pmatrix}$, $z = 1 - t_3$

$\therefore \begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \begin{pmatrix}y\\x\\z\end{pmatrix}$

5. fisheye projection

(equidistant fisheye projection)

5.3 $(x,y,z) \to (u,v)$

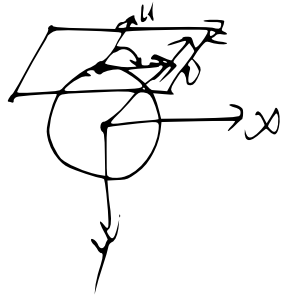
solve: $\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \left\|\begin{pmatrix}x\\y\\z\end{pmatrix}\right\|^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\therefore \begin{pmatrix}u\\v\end{pmatrix} = \begin{pmatrix}\tan^{-1}\left(\frac{x}{z}\right)\\ \pi - \cos^{-1} y\end{pmatrix}$

$\therefore \begin{pmatrix}u\\v\end{pmatrix} \leftarrow \begin{pmatrix}v\,cu\\ v\,su\end{pmatrix}$

$\therefore \begin{pmatrix}u\\v\end{pmatrix} \leftarrow r\begin{pmatrix}u\\v\end{pmatrix}$

5.3.2



$\therefore \theta = a\cdot\tan^{-1}\left(\frac{u}{a}\right)$

$\phi = \tan^{-1}\left(\frac{v\cos\theta}{b}\right)$

$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} c\phi s\theta \\ s\phi \\ c\phi c\theta \end{pmatrix}$

7.5 $\because f=1$, (see 1.7)

$\therefore$ projection plane is $z=1$.

7.6 $\because k\tan x > \tan(kx)$, $0<k<1$

$\therefore \tan x > a\tan\left(\frac{x}{a}\right)$, $a>1$

7.7 $\because c^2x = \frac{c^2}{s^2+c^2} = \frac{1}{t^2+1}$

$\therefore \tan^2 x = \frac{1}{\cos^2 x} - 1$

7.8 $\tan x > x$, $x\in(0,\frac{\pi}{2})$

proof: let $f(x) = tx - x$

$\because f'(x) = \frac{1}{c^2x} - 1 = t^2x > 0$, $x\in(0,\frac{\pi}{2})$

$\because f(0)=0$ $\therefore f(x)>0$ for $x\in(0,\frac{\pi}{2})$

6.6 tangent half-angle identity

$t\frac{\alpha}{2} = \frac{s\alpha}{1+c\alpha} = \frac{1-c\alpha}{s\alpha}$

7. compressed rectilinear projection

7.3 $(x,y,z)\to(u,v)$

solve: $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \leftarrow \left\|\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right\|^{-1}\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\begin{pmatrix} \theta \\ \phi \end{pmatrix} = \begin{pmatrix} \tan2^{-1}(x,z) \\ \sin^{-1}(y) \end{pmatrix}$

$\therefore \begin{pmatrix} u \\ v \end{pmatrix} = r\begin{pmatrix} a\tan\frac{\theta}{a} \\ b\tan\phi/\cos\theta \end{pmatrix}$

here $\theta, \frac{\theta}{a}, \phi \in (-\frac{\pi}{2}, \frac{\pi}{2})$

(see 1.7)

7.3.2 $\therefore \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \to \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$

7.4 $(u,v)\to(x,y,z)$

solve: $\begin{pmatrix} y \\ v \end{pmatrix} \leftarrow r^{-1}\begin{pmatrix} u \\ v \end{pmatrix}$

solve: $\begin{pmatrix} y \\ v \end{pmatrix} \leftarrow r^{-1}\begin{pmatrix} u \\ v \end{pmatrix}$, $r_\beta = \left\|\begin{pmatrix} y \\ v \end{pmatrix}\right\|$

$\begin{pmatrix} u \\ \alpha \end{pmatrix} \leftarrow \begin{pmatrix} \tan2^{-1}(v,u) \\ \pi - 2\tan^{-1}(r_\beta^{-1}) \end{pmatrix}$

$v \leftarrow \alpha$

$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} svsu \\ cv \\ svcu \end{pmatrix}$

6.4.2 $\alpha = \pi - 2\tan^{-1}(r_\beta^{-1})$

proof: let $\alpha = \pi - 2\beta$,

$\therefore \beta = \frac{\pi}{2} - \frac{\alpha}{2}$

$\therefore \tan\beta = \frac{2}{r_p} = r_\beta^{-1} \in (0,+\infty]$

$\therefore \beta = \tan^{-1}(r_\beta^{-1}) \in (0,\frac{\pi}{2}]$

$\therefore \alpha = \pi - 2\beta = \pi - 2\tan^{-1}(r_\beta^{-1})$

$\in [0,\pi)$

6.5 projection plane $Y=1$, projection pole $=(0,-1,0)$

8. recti-perspective projection (Panini projection)

$\therefore \begin{pmatrix}\theta\\ \phi\end{pmatrix} = \begin{pmatrix}\tan^{-1}(x,z)\\ \sin^{-1} y\end{pmatrix}$

$\therefore \begin{pmatrix}u\\ v\end{pmatrix} = \begin{pmatrix}u'\\ v'\end{pmatrix} = r\begin{pmatrix}a\tan\frac{\theta}{a}\\ b\frac{\tan\phi}{\cos\theta}\end{pmatrix}$

8.2

8.2.1 rectilinear projection (see 1.7)

$\begin{pmatrix}u\\ v\end{pmatrix} = \begin{pmatrix}\tan\theta\\ \tan\phi/\cos\theta\end{pmatrix}$

$\theta, \phi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\therefore \begin{pmatrix}u\\ v\end{pmatrix} \leftarrow \begin{pmatrix}-u\\ v\end{pmatrix}$

7.b.3.2

$\therefore \begin{pmatrix}1&2\\3&4\\5&6\end{pmatrix} \rightarrow \begin{array}{c}\rightarrow v'\\ \downarrow u'\end{array}\begin{pmatrix}1&2\\3&4\\5&6\end{pmatrix}$

8.2.2 compressed rectilinear projection

$\begin{pmatrix}u\\ v\end{pmatrix} = \begin{pmatrix}a\tan\frac{\theta}{a}\\ b\tan\phi/\cos\theta\end{pmatrix}$ (see 7.3)

$\theta, \frac{\theta}{a}, \phi \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\therefore \begin{pmatrix}1&2\\3&4\\5&6\end{pmatrix} \rightarrow \begin{pmatrix}5&3&1\\6&4&2\end{pmatrix}$

7.b.4 $(u,v) \rightarrow (x,y,z)$

solve: $\begin{pmatrix}u\\ v\end{pmatrix} \leftarrow \begin{pmatrix}-u\\ v\end{pmatrix}$, $\begin{pmatrix}u\\ v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\ v\end{pmatrix}$

8.2.3 recti-perspective projection

① $\begin{pmatrix}u\\ v\end{pmatrix} = \begin{pmatrix}a\tan\frac{\theta}{a}\\ b\frac{v_0}{u_0}u\end{pmatrix}$, ($u_0, v_0$ is u, v in 8.2.1)

$\therefore \begin{pmatrix}u\\ v\end{pmatrix} = \begin{pmatrix}a\tan\frac{\theta}{a}\\ b\frac{\tan\phi}{\cos\theta}\frac{1}{\tan\theta}a\tan\frac{\theta}{a}\end{pmatrix}$

$\therefore \theta = a\cdot\tan^{-1}\frac{u}{a}$,

$\phi = \tan^{-1}\frac{v\cos\theta}{b}$

$\therefore \begin{pmatrix}x\\ y\\ z\end{pmatrix} = \begin{pmatrix}c\phi s\theta\\ s\phi\\ c\phi c\theta\end{pmatrix}$

$\therefore \begin{pmatrix}x\\ y\\ z\end{pmatrix} \leftarrow \begin{pmatrix}y\\ x\\ z\end{pmatrix}$

7.9 $k\tan(x) > \tan(kx)$, $0<k<1$

$k\tan(x) < \tan(kx)$, $k>1$

here $x \in \left(0, \frac{\pi}{2}\right)$, $kx \in \left(0, \frac{\pi}{2}\right)$, $k>0$

proof: let $f(x) = t(kx) - kt(x)$

$\therefore f'(x) = \frac{k}{c^2(kx)} - k\frac{1}{c^2 x}$

$= k\left(\frac{1}{c^2(kx)} - \frac{1}{c^2(x)}\right)$

$\therefore f'(x) < 0$, $0<k<1$

$f'(x) > 0$, $k>1$

$\because f(0) = 0$, $\therefore f(x) < 0$, $0<k<1$

$f(x) > 0$, $k>1$

7.b compressed rectilinear portrait projection

7.b.3 $(x,y,z) \rightarrow (u,v)$

(see 2.b.3)

solve: $\begin{pmatrix}x\\ y\\ z\end{pmatrix} \leftarrow \begin{pmatrix}y\\ x\\ z\end{pmatrix}$, $\begin{pmatrix}x\\ y\\ z\end{pmatrix} \leftarrow \left\|\begin{pmatrix}x\\ y\\ z\end{pmatrix}\right\|^{-1}\begin{pmatrix}x\\ y\\ z\end{pmatrix}$

8.b.4 $(u,v)\to(x,y,z)$

solve: $\begin{pmatrix}y\\v\end{pmatrix}\leftarrow\begin{pmatrix}-u\\v\end{pmatrix}$, $\begin{pmatrix}y\\v\end{pmatrix}\leftarrow r^{-1}\begin{pmatrix}y\\v\end{pmatrix}$

$$\theta=a\tan^{-1}\frac{u}{a}$$

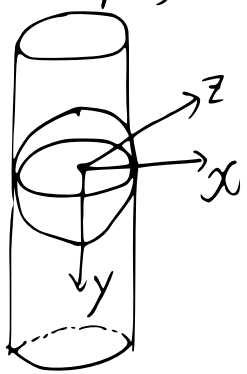
$$\phi=\begin{cases}\tan^{-1}\left(\dfrac{v\sin\theta}{ab\tan\frac{\theta}{a}}\right), & \text{if } \theta\neq0\\ \tan^{-1}\left(\frac{v}{b}\right), & \text{if } \theta=0\end{cases}$$

$$\therefore\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}c\phi s\theta\\ s\phi\\ c\phi c\theta\end{pmatrix}$$

$$\therefore\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow\begin{pmatrix}y\\x\\z\end{pmatrix}$$

9. Mercator projection

9.2



8.4 $(u,v)\to(x,y,z)$

solve: $\begin{pmatrix}y\\v\end{pmatrix}\leftarrow r^{-1}\begin{pmatrix}y\\v\end{pmatrix}$

$$\theta=a\tan^{-1}\frac{u}{a}$$

$$\phi=\begin{cases}\tan^{-1}\left(\dfrac{v\sin\theta}{ab\tan\frac{\theta}{a}}\right), & \text{if } \theta\neq0\\ \tan^{-1}\left(\frac{v}{b}\right), & \text{if } \theta=0\end{cases}$$

$$\therefore\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}c\phi s\theta\\ s\phi\\ c\phi c\theta\end{pmatrix}\quad\text{(see 7.4)}$$

8.b Panini portrait projection

8.b.3 $(x,y,z)\to(u,v)$

solve: $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow\begin{pmatrix}y\\x\\z\end{pmatrix}$, $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow\left\|\begin{pmatrix}x\\y\\z\end{pmatrix}\right\|^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\theta=\tan2^{-1}(x,z)$, $\phi=\sin^{-1}y$ (see 7.b.3)

$$\therefore\ u=a\tan\frac{\theta}{a}$$

$$v=\begin{cases}b\tan\phi, & \text{if } \theta=0\\ bu\tan\phi/\sin\theta, & \text{if } \theta\neq0\end{cases}$$

$\therefore\begin{pmatrix}y\\v\end{pmatrix}\leftarrow r\begin{pmatrix}y\\v\end{pmatrix}$, $\therefore\begin{pmatrix}y\\v\end{pmatrix}\leftarrow\begin{pmatrix}-u\\v\end{pmatrix}$

$$\therefore\begin{pmatrix}y\\v\end{pmatrix}=\begin{pmatrix}a\tan\frac{\theta}{a}\\ ab\dfrac{\tan\phi}{\sin\theta}\tan\frac{\theta}{a}\end{pmatrix}$$

here $\theta\neq0$, $\theta,\frac{\theta}{a},\phi\in(-\frac{\pi}{2},\frac{\pi}{2})$

② $\therefore\begin{pmatrix}y\\v\end{pmatrix}:=\begin{pmatrix}a\tan\frac{\theta}{a}\\ ab\dfrac{\tan\phi}{\sin\theta}\tan\frac{\theta}{a}\end{pmatrix}$, $\theta\neq0$

$$\because\lim_{\theta\to0}\frac{a\tan\frac{\theta}{a}}{\sin\theta}=1$$

$$\therefore\begin{pmatrix}y\\v\end{pmatrix}:=\begin{pmatrix}a\tan\frac{\theta}{a}\\ b\tan\phi\end{pmatrix},\ \text{if } \theta=0$$

here $\frac{\theta}{a},\phi\in(-\frac{\pi}{2},\frac{\pi}{2})$, $\theta\in(-\pi,\pi)$

8.3 $(x,y,z)\to(u,v)$

solve: $\begin{pmatrix}x\\y\\z\end{pmatrix}\leftarrow\left\|\begin{pmatrix}x\\y\\z\end{pmatrix}\right\|^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$

$$\begin{pmatrix}\theta\\\phi\end{pmatrix}=\begin{pmatrix}\tan2^{-1}(x,z)\\ \sin^{-1}(y)\end{pmatrix}$$

$$\therefore\begin{pmatrix}y\\v\end{pmatrix}=r\begin{pmatrix}a\tan\frac{\theta}{a}\\ ab\dfrac{\tan\phi}{\sin\theta}\tan\frac{\theta}{a}\end{pmatrix},\ \text{if } \theta\neq0$$

$$\begin{pmatrix}y\\v\end{pmatrix}=r\begin{pmatrix}a\tan\frac{\theta}{a}\\ b\tan\phi\end{pmatrix},\ \text{if } \theta=0$$

$\therefore \frac{dv}{d\phi} = \frac{1}{\cos\phi}$

$\therefore v = \int \frac{1}{\cos\phi} d\phi$

$= \int \sec\phi \, d\phi$

$= \ln(\tan(\frac{\pi}{4}+\frac{\phi}{2})) + C$

$\therefore v(\phi) = \ln(\tan(\frac{\pi}{4}+\frac{\phi}{2}))$

9.9 $v(\phi) = \ln(\tan(\frac{\pi}{4}+\frac{\phi}{2})) \Rightarrow v'(\phi) = \sec(\phi)$

proof: let $\alpha = \frac{\pi}{4}+\frac{\phi}{2}$,

$\therefore v' = \frac{1}{t\alpha}\sec^2(\alpha)\cdot\frac{1}{2}$

$= \frac{1}{2}\frac{c\alpha}{s\alpha}\frac{1}{c^2\alpha}$

$= \frac{1}{2s\alpha c\alpha} = \frac{1}{s(2\alpha)}$

$= \frac{1}{s(\frac{\pi}{2}+\phi)} = \frac{1}{c\phi}$

$= \sec\phi$

9.6 $\cosh(x) = \frac{e^x+e^{-x}}{2}$

$\sinh(x) = \frac{e^x-e^{-x}}{2}$

9.7 $e^v = \tan(\frac{\pi}{4}+\frac{\phi}{2}) \Rightarrow \sinh(v) = \tan\phi$

proof: $\sinh(v) = \frac{e^v-e^{-v}}{2}$

$= \frac{1}{2}(e^v - \frac{1}{e^v})$

$\because e^v = \frac{1+t\frac{\phi}{2}}{1-t\frac{\phi}{2}}$

$\therefore \sinh(v) = \frac{1}{2}(\frac{1+t\frac{\phi}{2}}{1-t\frac{\phi}{2}} - \frac{1-t\frac{\phi}{2}}{1+t\frac{\phi}{2}})$

$= \frac{1}{2}\frac{4t\frac{\phi}{2}}{1-t^2\frac{\phi}{2}}$

$= t\phi$

9.8 ① horizontal scale factor

$k_h(\phi) = \frac{2\pi}{2\pi\cos\phi} = \frac{1}{\cos\phi}$

② $\because$ vertical scale factor

$k_v(\phi) = k_h(\phi)$

9.3 $(x,y,z) \to (u,v)$

solve: $\begin{pmatrix}x\\y\\z\end{pmatrix} \leftarrow \left\|\begin{pmatrix}x\\y\\z\end{pmatrix}\right\|^{-1}\begin{pmatrix}x\\y\\z\end{pmatrix}$

$\begin{pmatrix}\theta\\\phi\end{pmatrix} = \begin{pmatrix}\tan^{-1}(x,z)\\ \sin^{-1}y\end{pmatrix}$

$\therefore \begin{pmatrix}u\\v\end{pmatrix} = \begin{pmatrix}\theta\\ gd^{-1}(\phi) = \frac{1}{2}\ln\frac{1+s\phi}{1-s\phi} = \frac{1}{2}\ln\frac{1+y}{1-y}\end{pmatrix}$

$\begin{pmatrix}u\\v\end{pmatrix} \leftarrow r\begin{pmatrix}u\\v\end{pmatrix}$

(note $gd^{-1}(\phi) = \ln\tan(\frac{\pi}{4}+\frac{\phi}{2})$)

9.4 $(u,v) \to (x,y,z)$

solve: $\begin{pmatrix}u\\v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\v\end{pmatrix}$

$\begin{pmatrix}\theta\\\phi\end{pmatrix} = \begin{pmatrix}u\\ gd(v)\end{pmatrix} = \begin{pmatrix}u\\ \tan^{-1}(\sinh(v))\end{pmatrix}$

$\therefore \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}c\phi s\theta\\ s\phi\\ c\phi c\theta\end{pmatrix}$

9.5 $\because \phi \in (-\frac{\pi}{2},\frac{\pi}{2})$ $\therefore \frac{1}{2}(\frac{\pi}{2}+\phi) \in (0,\frac{\pi}{2})$

$\therefore \tan(\frac{1}{2}(\frac{\pi}{2}+\phi)) > 0$,

$\therefore \ln\tan(\frac{1}{2}(\frac{\pi}{2}+\phi))$ is defined.

10.5 $(u,v) \to (\theta,\phi)$

solve: $\because \begin{pmatrix}\theta\\ \phi\end{pmatrix} = \begin{pmatrix}\tan2^{-1}(x,z)\\ \sin^{-1}y\end{pmatrix}$

$$= \begin{pmatrix}\tan2^{-1}(\tan\phi', \cos\theta')\\ \sin^{-1}(\cos\phi'\sin\theta')\end{pmatrix}$$

$\because \cos\phi' = \operatorname{sech}(u)$

$\tan\phi' = \sinh(u)$

$\therefore \begin{pmatrix}\theta\\ \phi\end{pmatrix} = \begin{pmatrix}\tan2^{-1}(\sinh(u), \cos v)\\ \sin^{-1}(\sin(v)/\cosh(u))\end{pmatrix}$

11. affine homography

11.2 $K(R\ t) = K(I\ 0)$,

$K =$ affine $H \in R^{3\times3}$

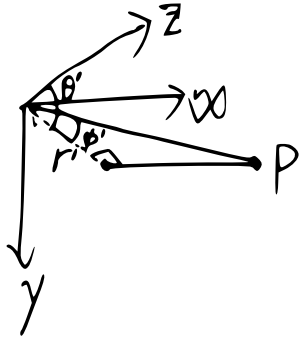
$H_w = H^{-1} = K^{-1}$

11.2.3 $p \to p_w \in R^2$

solve: $\begin{pmatrix}p_w\\ 1\end{pmatrix} = H_w\begin{pmatrix}p\\ 1\end{pmatrix} = K^{-1}\begin{pmatrix}p\\ 1\end{pmatrix}$

11.2.4 $p_w \to p \in R^2$

solve: $\begin{pmatrix}p\\ 1\end{pmatrix} = H\begin{pmatrix}p_w\\ 1\end{pmatrix} = K\begin{pmatrix}p_w\\ 1\end{pmatrix}$



$$\begin{cases} x = s\phi' \\ r' = c\phi' \\ y = r's\theta' \\ z = r'c\theta' \end{cases}$$

10.3 $(x,y,z) \to (u,v)$

solve: $\begin{pmatrix}x\\ y\\ z\end{pmatrix} \leftarrow \left\|\begin{pmatrix}x\\ y\\ z\end{pmatrix}\right\|^{-1}\begin{pmatrix}x\\ y\\ z\end{pmatrix}$

$\begin{pmatrix}\theta'\\ \phi'\end{pmatrix} = \begin{pmatrix}\tan2^{-1}(y,z)\\ \sin^{-1}x\end{pmatrix}$

$\begin{pmatrix}u\\ v\end{pmatrix} = \begin{pmatrix}gd^{-1}(\phi') = \frac{1}{2}\ln\frac{1+s\phi'}{1-s\phi'} = \frac{1}{2}\ln\frac{1+x}{1-x}\\ \theta'\end{pmatrix}$

$\begin{pmatrix}u\\ v\end{pmatrix} \leftarrow r\begin{pmatrix}u\\ v\end{pmatrix}$

10.3.2 $\therefore \begin{pmatrix}1&2\\3&4\\5&6\end{pmatrix} \to \begin{pmatrix}1&2\\3&4\\5&6\end{pmatrix}$

10.4 $(u,v) \to (x,y,z)$

solve: $\begin{pmatrix}u\\ v\end{pmatrix} \leftarrow r^{-1}\begin{pmatrix}u\\ v\end{pmatrix}$

$\begin{pmatrix}\theta'\\ \phi'\end{pmatrix} = \begin{pmatrix}v\\ gd(u)\end{pmatrix} = \begin{pmatrix}v\\ \tan^{-1}\sinh(u)\end{pmatrix}$

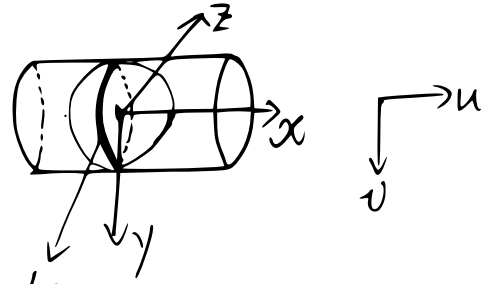
$\therefore \begin{pmatrix}x\\ y\\ z\end{pmatrix} = \begin{pmatrix}s\phi'\\ c\phi's\theta'\\ c\phi'c\theta'\end{pmatrix}$

10. transverse Mercator projection

(spherical TM projection,

Gauss projection)

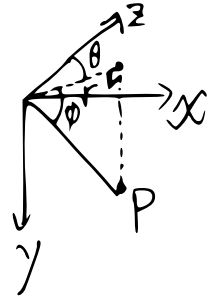
10.2



transverse equator plane

10.2.1 for Mercator projection,

P is projected onto plane $Y=0$ first,



$$\begin{cases} y = s\phi \\ r = c\phi \\ x = rs\theta \\ z = rc\theta \end{cases}$$

10.2.2 for TM projection,

P is projected onto plane $X=0$ first,

② for plane projection:

$p \to (x,y,z) \to p_w^T = (u,v)$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = R^{-1}K^{-1}\begin{pmatrix} p \\ 1 \end{pmatrix}$,

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \leftarrow z^{-1}(1-t_3)\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} t_1 \\ t_2 \end{pmatrix}$

$\because K=K',\ R=R',\ t=t'$

$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} A^{-1} & \\ & 1 \end{pmatrix}\begin{pmatrix} p \\ 1 \end{pmatrix}$

$\therefore \begin{pmatrix} p_w \\ 1 \end{pmatrix} = \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} + \begin{pmatrix} t_1 \\ t_2 \\ 0 \end{pmatrix}$

$= \begin{pmatrix} A^{-1} & \\ & 1 \end{pmatrix}\begin{pmatrix} p \\ 1 \end{pmatrix} + \begin{pmatrix} -A^{-1}b \\ 0 \end{pmatrix}$

$= H^{-1}\begin{pmatrix} p \\ 1 \end{pmatrix} = H_w\begin{pmatrix} p \\ 1 \end{pmatrix}$

proved.

11.6 kludgy method

$K = H = \begin{pmatrix} A & b \\ & 1 \end{pmatrix} \in R^{3\times3}$, $\begin{pmatrix} p \\ 1 \end{pmatrix} = H\begin{pmatrix} p_w \\ 1 \end{pmatrix}$

let $K'=1$, $R' = \begin{pmatrix} A & \\ & 1 \end{pmatrix}$, $t' = \begin{pmatrix} -A^{-1}b \\ 0 \end{pmatrix}$

(note R' is not rotation matrix)

$\Rightarrow$ plane projection with $K=K'$,

$R=R'$, $t=t'$ is affine homography.

(see 4)

proof:

① $\begin{pmatrix} p \\ 1 \end{pmatrix} = H\begin{pmatrix} p_w \\ 1 \end{pmatrix} = \begin{pmatrix} A & b \\ & 1 \end{pmatrix}\begin{pmatrix} p_w \\ 1 \end{pmatrix}$

$\therefore \begin{pmatrix} p \\ 1 \end{pmatrix} = \begin{pmatrix} A & \\ & 1 \end{pmatrix}\begin{pmatrix} p_w \\ 1 \end{pmatrix} + \begin{pmatrix} 0 & b \\ & 0 \end{pmatrix}\begin{pmatrix} p_w \\ 1 \end{pmatrix}$

$= \begin{pmatrix} A & \\ & 1 \end{pmatrix}\begin{pmatrix} p_w \\ 1 \end{pmatrix} + \begin{pmatrix} b \\ 0 \end{pmatrix}$

$\because H_w = H^{-1} = \begin{pmatrix} A' & b' \\ & 1 \end{pmatrix} = \begin{pmatrix} A^{-1} & -A^{-1}b \\ & 1 \end{pmatrix} \in R^{3\times3}$

$\therefore \begin{pmatrix} p_w \\ 1 \end{pmatrix} = H_w\begin{pmatrix} p \\ 1 \end{pmatrix} = \begin{pmatrix} A' & \\ & 1 \end{pmatrix}\begin{pmatrix} p \\ 1 \end{pmatrix} + \begin{pmatrix} b' \\ 0 \end{pmatrix}$

11.3 $(x,y,z) \to (u,v) = p_w^T$

solve: $\because \begin{pmatrix} x \\ y \\ z \end{pmatrix} = R^{-1}K^{-1}\begin{pmatrix} p \\ 1 \end{pmatrix}$

互 $\because K(R\ t) = K(I\ 0)$

$\therefore \begin{pmatrix} x \\ y \\ z \end{pmatrix} = K^{-1}\begin{pmatrix} p \\ 1 \end{pmatrix} = H_w\begin{pmatrix} p \\ 1 \end{pmatrix} = \begin{pmatrix} p_w \\ 1 \end{pmatrix}$

$\therefore \begin{pmatrix} u \\ v \end{pmatrix} = p_w = \begin{pmatrix} x \\ y \end{pmatrix}$

11.4 $p_w^T = (u,v) \to (x,y,z)$

solve: $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} p_w \\ 1 \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix}$ (from 11.3)

11.5 $\forall$ affine $H = \begin{pmatrix} A & b \\ & 1 \end{pmatrix}$, A^{-1} exists,

$H \in R^{(n+1)\times(n+1)}$, $n \geq 1$

$\begin{pmatrix} x' \\ 1 \end{pmatrix} = H\begin{pmatrix} x \\ 1 \end{pmatrix} \Rightarrow H^{-1} = \begin{pmatrix} A^{-1} & -A^{-1}b \\ & 1 \end{pmatrix}$

proof: $\because x' = Ax + b$

$\therefore x = A^{-1}(x'-b) = A^{-1}x' - A^{-1}b$

$\therefore \begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} A^{-1} & -A^{-1}b \\ & 1 \end{pmatrix}\begin{pmatrix} x' \\ 1 \end{pmatrix}$

$\therefore H^{-1} = \begin{pmatrix} A^{-1} & -A^{-1}b \\ & 1 \end{pmatrix}$